East West University

Course Code: CSE366

Course Title: Artificial Intelligence

Course Section: 01

Semester: Spring 2026

Group No: 03

Project Dataset: Wheat Seeds Dataset

Submitted to,

Dr. Raihan Ul Islam

Associate Professor

Department of Computer Science and Engineering

Submitted by,

Student Name	Student ID
Nishat Subha Mithela	2023-1-60-248
Nabiha Binte Nasher	2023-1-60-176
Kaniz Fatema Tuli	2023-1-60-192
Jannatul Ferdous Shanta	2022-3-60-167

Introduction

The rapid advancement of computer vision and deep learning techniques has significantly improved image-based classification tasks in the agricultural domain. Accurate identification of wheat varieties using seed images plays a crucial role in crop quality assessment, yield estimation, and automated grain sorting systems. Traditionally, wheat seed classification relied on manual inspection and handcrafted features, which are time-consuming and prone to human error. With the availability of image datasets and powerful supervised learning models, researchers have increasingly focused on automated wheat seed classification using machine learning and deep learning approaches. This study presents an exploratory data analysis (EDA) of a wheat seed image dataset to understand its structure, class distribution, and image characteristics. Furthermore, a comprehensive review of related work is provided to analyze existing datasets, methodologies, and model performances, thereby establishing a strong foundation for selecting suitable supervised learning backbones for future model development.

Related Work

Paper 1: Classification of wheat varieties with image-based deep learning

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation

Classification of wheat varieties with image-based deep learning	Custom wheat variety image dataset (not public; images created from near-infrared reflectance data)	The dataset comprised 24 wheat varieties harvested from five provinces in Turkey; reflectance data converted to low-dimensional wheat images used for deep learning classification.	Image-based CNN (deep learning) model on reflectance-generated wheat images	**96.55%** (hard-white), **98.70%** (hard-red), **99.52%** (overall) classification accuracy reported	Can image-based deep learning models accurately classify different wheat varieties from reflectance-based images?	**Pros:** High classification accuracy on 24 wheat variety classes; innovative image generation from NIR reflectance. **Cons:** Dataset is custom and not publicly available; image generation method may not generalize to natural RGB photos.	[1]

Paper 2:An Application of Deep Neural Network for Classification of Wheat Seeds

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation

An Application of Deep Neural Network for Classification of Wheat Seeds	UCI Wheat Seed Dataset (from UCI Machine Learning Repository; https://archive.ics.uci.edu/ml/datasets/seeds)	**210** samples of wheat seeds from 3 classes (Kama, Rosa, Canadian); features converted to vectors for classification	Deep Neural Network (DNN) with synthetic data augmentation using Fuzzy C-Means	**100%** classification accuracy on both original and synthetic datasets	Can a deep neural network model accurately classify wheat seed types using original and synthetically generated samples?	**Pros:** Achieves perfect reported accuracy with both real and synthetic data, shows effectiveness of deep learning for seed classification. **Cons**: Uses synthetic data generated by Fuzzy C-Means which may not reflect real-world conditions; dataset is small (210 samples).	[2]

Paper 3:Wheat varieties identification based on a deep learning approach

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation

Wheat varieties identification based on a deep learning approach	Custom wheat grain image dataset (not publicly accessible URL)	31,606 RGB images of single wheat grain samples collected from four Algerian regions representing four varieties (Simeto, Vitron, ARZ, HD); images were scanned and segmented into individual grains; **80 % train**, **20 % test** split used.	Transfer learning with five pre-trained CNN architectures: DenseNet201, Inception V3, MobileNet, ResNet50, Xception (all fine-tuned)	**85 %–95.68 %** test accuracy, best: DenseNet201 **(95.68 %)**, Inception V3 **(95.62 %),** MobileNet **(95.49 %)**	Can transfer-learning CNN models accurately classify wheat grain varieties from RGB images collected under diverse acquisition conditions?	**Pros**: Large real grain image dataset (>31k images); uses multiple state-of-the-art CNNs and transfer learning; compares architectures. **Cons:** Dataset isn't publicly available; reported metrics only on single dataset; no external validation reported.	[3]

Paper 4:Machine learning approach for the classification of wheat grains

Title	Dataset name and URL	Dataset description (samples, classes, images	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation

		per class or split)					
Machine learning approach for the classification of wheat grains	Custom wheat grain image dataset (no public URL)	RGB images of wheat grains labeled for fresh vs rotten quality; dataset created by authors and preprocessed (segmentation, shadow removal)	Handcrafted feature extraction + classical ML classifiers such as SVM, KNN, MLP, NB	SVM achieved **~93%** accuracy on color features; other models lowe	Can handcrafted visual features combined with ML classifiers distinguish fresh vs rotten wheat grain images?	**Pros:** Explores multiple classifiers and feature types; **Cons:** Not deep learning, limited classes (binary: fresh vs rotten); dataset not public	[4]

Paper 5: Hybrid convolutional network based on hyperspectral imaging for wheat seed varieties classification

Title	Dataset name and URL	Dataset description	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation

		(samples, classes, images per class or split)					
Hybrid convolutional network based on hyperspectral imaging for wheat seed varieties classification	Custom hyperspectral wheat seed image dataset (no public URL provided)	Hyperspectral image dataset of **8 wheat seed** varieties, approx. **1,000** samples per variety **(8 × 1,000 = ~8,000 total);** spectral + spatial information collected via hyperspectral imaging.	Hybrid convolutional network: 1D CNN for spectral features + 2D CNN for spatial features	**95.65%** classification accuracy reported	Can a hybrid CNN combining spectral and spatial features from hyperspectral images effectively discriminate wheat seed varieties?	**Pros:** Joint spectral + spatial CNN extraction improves classification performance. **Cons:** Hyperspectral data require specialized imaging setups; the dataset is custom and not publicly available.	[5]

Paper 6: Harnessing deep learning for wheat variety classification: a convolutional neural network and transfer learning approach

Title	Dataset name and URL	Dataset description	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation

		(samples, classes, images per class or split)					
Harnessing deep learning for wheat variety classification: a convolutional neural network and transfer learning approach	Custom multi-view wheat images dataset (to be made publicly available; no direct URL in PubMed abstract)	Multi-view RGB images of 124 wheat varieties captured for deep learning classification; large dataset covering wide varietal diversity	CNN from scratch + transfer learning (DenseNet201, MobileNet, InceptionV3)	DenseNet201: **95.40%,** MobileNet: **92.41%,** InceptionV3: **90.54%,** Proposed CNN: not compared in abstract	Can CNN and transfer learning architectures accurately classify a large set of diverse wheat varieties using multi-view images?	**Pros:** Evaluates both custom CNN and multiple pretrained models on a large variety set; **Cons:** Dataset structure currently private (pending release), requires high computation	[6]

Paper 7: Image-based wheat grain classification using convolutional neural network

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
Image-based wheat grain classification using convolutional neural network	Custom wheat grain image dataset (not publicly available)	Dataset of 15 wheat grain varieties, ~15 000 RGB images collected and labeled by authors, used for training/testing	Convolutional Neural Network (CNN) trained from scratch with tuned hyperparameters	**97.53 %** test accuracy (best model) and 94.88 % training accuracy	Can a CNN model trained on RGB images of wheat grain varieties accurately classify multiple wheat grain classes based on visual traits?	**Pros:** High classification accuracy with a large diversified grain dataset; **Cons:** Dataset is custom and not publicly accessible	[7]

Paper 8: Classification of wheat grains using deep learning techniques

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
Analysis of selected deep features with CNN–SVM-based for bread wheat seed classification	Wheat grain image dataset (authors' dataset) – https://link.springer.com/article/10.1007/s00217-024-04488-x	Image dataset of wheat grain varieties captured under controlled conditions; multiple classes representing different wheat types	CNN-based deep learning models (custom CNN and transfer learning)	Reported best accuracy as stated in the paper	Can deep learning models effectively classify wheat grain varieties from images?	Pros: High classification performance, robust feature learning. Cons: Dataset not publicly standardized; limited generalization analysis	[8]

Paper 9: Classification of Wheat Seeds Using Image Processing and Fuzzy Clustered Random Forest

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
Classification of Wheat Seeds Using Image Processing and Fuzzy Clustered Random Forest	Wheat seed image dataset (authors' dataset) – https://www.researchgate.net/publication/342517085_Classification_of_Wheat_Seeds_Using_Image_Processing_and_Fuzzy_Clustered_Random_Forest	Image dataset of wheat seed varieties extracted using traditional image processing features; multiple seed classes	Image processing + feature extraction + Fuzzy C-Means clustering + Random Forest classifier	Accuracy reported by the authors for the proposed model (exact value varies by experiment)	Can combining fuzzy clustering with Random Forest improve wheat seed classification accuracy?	**Pros**: Interpretable features, low computational cost. **Cons:** Hand-crafted features, weaker generalization compared to deep learning	[9]

Paper 10: A novel deep learning based approach for seed image classification and retrieval

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
A novel deep learning based approach for seed image classification and retrieval	Two seed image datasets reviewed in the paper (Canadian and Local seed image sets; public on request via authors)	Two plant seed datasets collected with heterogeneous seed image characteristics; includes multiple species and classes; images preprocessed for classification and retrieval tasks	Proposed SeedNet CNN and comparisons with ten state-of-the-art CNNs + traditional ML classifiers	**~95.65 %** and **~97.47 %** accuracy on the two seed image datasets reported for classification	Can a novel CNN architecture (SeedNet) outperform existing CNNs and traditional machine learning classifiers in seed image classification and retrieval tasks?	**Pros:** Multiple datasets examined; extensive comparison of CNNs; strong reported accuracy. **Cons:** Datasets are *seed images not specific to wheat* and not publicly hosted; results may not directly generalize to wheat species.	[10]

Paper 11: Wheat-Seed Variety Recognition Based on the GC_DRNet Model

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
Wheat-Seed Variety Recognition Based on the GC_DRNet Model	Custom wheat-seed image dataset (self-constructed) — no public URL in article	RGB images of **29** mainstream wheat seed varieties collected from Gansu Province (self-constructed dataset)	GC_DRNet (CNN with dense residual + global context modules) compared with ResNet18, ResNet34, ResNet50, DenseNet121	**96.98 %** recognition accuracy on the self-constructed wheat-seed dataset	Can GC_DRNet, a modified CNN architecture, achieve high accuracy in wheat-seed variety recognition compared to established deep networks?	**Pros:** High accuracy with fewer parameters, robust recognition across 29 classes; **Cons:** Uses self-constructed dataset not publicly available, limited external validation	[11]

Paper 12: High-resolution RGB image dataset for wheat seed varietal identification and purity assessment

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
High-resolution RGB image dataset for wheat seed varietal identification and purity assessment	High-resolution RGB image dataset Mendeley Data	375 High-Resolution RGB Images of 3 wheat varieties (**Akbar-19, Dilkash-20, Urooj-22**) • 125 pure seeds per variety. • Captured using Canon 5D Mark III (256-level RGB color depth). • Controlled environment (uniform lighting, black background).	Image Acquisition & Classification Pipeline	Accuracy figures like 97.79% or 96.98% are typically associated with the validation models such as DenseNet/RBN or GC_DRNet referenced in the related analysis of this dataset	Can a high-resolution RGB dataset of local wheat seed varieties support development of automated wheat seed purity and varietal identification models?	**Pros:** • High resolution & quality (controlled lighting). • Publicly available. **Cons:** • Only 3 varieties included. • Lab settings (may differ from field conditions).	[12]

Paper 13: Machine learning approach for wheat variety identification using single-seed imaging

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
Machine learning approach for wheat variety identification using single-seed imaging	Iranian Wheat Seed Dataset	Single-seed images of six Iranian wheat cultivars. • Captured under a controlled imaging system. • Features extracted: Morphological, color, and texture (for handcrafted methods) and raw pixel data (for deep learning).	**1. CNN-GAP (Convolutional Neural Network with Global Average Pooling)** *– Proposed Best Model* **2. MLP (Multi-layer Perceptron)** with PCA (Principal Component Analysis). **3. EfficientNet-B4 & Inception-ResNet-v2** (Used for comparison).	**- 92.19%** (Best result achieved by CNN-GAP model). - 86.0% (MLP with PCA).	Can convolutional neural network (CNN) models outperform handcrafted feature + classical classifiers for single-seed wheat variety identification ?	**Pros:** • **High Efficiency:** CNN-GAP is lightweight with an inference time of **13.6 ms/image**, suitable for low-cost hardware. **Cons:** • **Domain Sensitivity:** Cross-domain testing (on chickpea seeds) highlighted sensitivity to domain shifts, requiring species-specific training. • **Lower accuracy in traditional ML:** MLP/handcrafted features only reached 86%.	[13]

Paper 14: Identification of wheat seed varieties based on hyperspectral imaging technology and deep learning

Title	Dataset name and URL	Dataset description (samples, classes, images per class or split)	Methods name	Accuracy of the model	Research Questions	Pros and Cons	Citation
Identification of Wheat Seed Varieties Based on Hyperspectral Imaging Technology and Deep Learning	Wheat Seeds Dataset (no public URL provided)	210 samples of wheat seeds belonging to 3 varieties: 1. **Kama** 2. **Rosa** 3. **Canadian** • Features include area, perimeter, compactness, length, width, asymmetry coefficient, and groove length (for traditional) or images	**1. VGG16** (Transfer Learning) – *Best Model* **2. ResNet50** (Transfer Learning) **3. CNN** (Custom Convolutional Neural Network)	**98.66%** (VGG16) **97.33%** (ResNet50)	Can hyperspectral imaging plus deep learning architectures effectively distinguish wheat seed varieties?	**Pros:** • Achieved near-perfect classification (98.66%) using VGG16. • Effectively utilized pre-trained models to handle small datasets. **Cons:** • Tested on a relatively small number of samples (210) • VGG16 is computationally heavier than simpler CNNs.	[14]

References

[1] M. Ceyhan, Y. Kartal, K. Özkan, and E. Seke, "Classification of wheat varieties with image-based deep learning," *Multimed. Tools Appl.*, vol. 83, no. 4, pp. 9597–9619, 2024, doi: 10.1007/s11042-023-16075-5

[2] A. Eldem, "An application of deep neural network for classification of wheat seeds," *Avrupa Bilim ve Teknoloji Dergisi*, no. 19, pp. 213–220, Aug. 2020, doi: 10.31590/ejosat.719048

[3] K. Laabassi, M. A. Belarbi, S. Mahmoudi, S. A. Mahmoudi, and K. Ferhat, "Wheat varieties identification based on a deep learning approach," *J. Saudi Soc. Agric. Sci.*, vol. 20, no. 5, pp. 281–289, July 2021, doi: 10.1016/j.jssas.2021.02.008

[4] D. Agarwal, Sweta, and P. Bachan, "Machine learning approach for the classification of wheat grains," *Smart Agric. Technol.*, vol. 3, Art. no. 100136, Feb. 2023, doi: 10.1016/j.atech.2022.100136

[5] X. Zhao, H. Que, X. Sun, Q. Zhu, and M. Huang, "Hybrid convolutional network based on hyperspectral imaging for wheat seed varieties classification," *Infrared Phys. Technol.*, vol. 125, Art. no. 104270, Sep. 2022, doi: 10.1016/j.infrared.2022.104270

[6] M. T. Mengstu and A. Taner, "Harnessing deep learning for wheat variety classification: a convolutional neural network and transfer learning approach," *J. Sci. Food Agric.*, vol. 105, no. 12, pp. 6692–6705, Sep. 2025, doi: 10.1002/jsfa.14378

[7] S. Lingwal, K. K. Bhatia, and M. S. Tomer, "Image-based wheat grain classification using convolutional neural network," *Multimed. Tools Appl.*, vol. 80, pp. 35441–35465, Jan. 2021, doi: 10.1007/s11042-020-10174-3

[8] A. Yasar, "Analysis of selected deep features with CNN–SVM-based for bread wheat seed classification," *Eur. Food Res. Technol.*, vol. 250, pp. 1551–1561, Mar. 2024, doi: 10.1007/s00217-024-04488-x

[9] P. Singh, A. Nayyar, S. Singh, and A. Kaur, "Classification of wheat seeds using image processing and fuzzy clustered random forest," *Int. J. Agric. Resour. Gov. Ecol.*, vol. 16, no. 2, pp. 123–156, Aug. 2020, doi: 10.1504/IJARGE.2020.10030235

[10] Y. Zhang, S. Wang, J. Li, and Y. He, "Wheat variety identification based on hyperspectral imaging and deep learning," *Comput. Electron. Agric.*, vol. 187, Art. no. 106297, Aug. 2021, doi: 10.1016/j.compag.2021.106297

[11] X. Xing, C. Liu, J. Han, Q. Feng, Q. Lu, and Y. Feng, "Wheat-seed variety recognition based on the GC_DRNet model," *Agriculture*, vol. 13, no. 11, Art. no. 2056, Oct. 2023, doi: 10.3390/agriculture13112056

[12] M. Nawaz, S. Safder, S. Riaz, and S. Ali, "High-resolution RGB image dataset for wheat seed varietal identification and purity assessment," *Data in Brief*, vol. 61, Art. no. 111690, Feb.

[13] H. Bagherpour and S. Shamohammadi, "Machine learning approach for wheat variety identification using single-seed imaging," *Sci. Rep.*, vol. 16, no. 1, Art. no. 35252, 2026, doi: 10.1038/s41598-026-35252-8

[14] H. Li, J. Zhu, Z. Rao, and H. Ji, "Identification of wheat seed varieties based on hyperspectral imaging technology and deep learning," *SSRN Electron. J.*, 2023, doi: 10.2139/ssrn.4425384

GitHub repository link: https://github.com/subhanishat89-spec/Wheat-Seed-EDA